

Brando Vasquez Rodriguez

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AVAILABLE 2nd CO-OP SPRING 2027 | 3rd CO-OP FALL 2027

EDUCATION

Bachelor of Science, Electrical Engineering

University of Cincinnati, Cincinnati, OH

Graduating: May 2029

GPA: 3.97

SKILLS

Software: Revit, AutoCAD, AGI32, C++, Python, MATLAB, SQL, LTspice, KiCad, LabVIEW, Verilog, Arduino IDE, Autodesk Fusion 360, Microsoft Office 365.

Hardware/Lab: Oscilloscope, Function Generator, Multimeter, Soldering, Embedded Systems, Sensor Integration.

Languages: English, Spanish.

PROFESSIONAL EXPERIENCE

Electrical Engineering Co-op

June 2026 - August 2026

IMEG, Blue Ash, OH

- Updated power, lighting, systems, and demolition drawings in Revit and AutoCAD across 10+ commercial and industrial projects, incorporating markups, circuiting, keynotes, and equipment changes.
- Completed 5 field surveys at P&G and GE Aerospace facilities, documenting electrical equipment for project drawings.
- Revised 8 single-line diagrams involving panels, transformers, disconnects, feeders, and transformer current calculations.

Undergraduate Research Assistant

October 2025 - May 2026

R.O.S.E Lab, University of Cincinnati

- Analyzed time-series physiological data using statistical methods to examine stress patterns in engineering students.
- Performed structured literature reviews and data organization to support experimental design and research methodology.
- Collaborated with faculty and interdisciplinary teams to maintain research documentation and data reproducibility.

PROJECTS & ADDITIONAL EXPERIENCE

Gym WRO - Wearable Rep Optimizer (Project)

May 2026 - July 2026

University of Cincinnati, Cincinnati, OH

- Led a 5-person multidisciplinary team developing a wearable system to track rep quality, range of motion, and fatigue.
- Built and programmed an ESP32 wearable from KiCad schematic through soldered hardware, integrating EMG/IMU sensing, battery power, and real-time RGB LED feedback.
- Evaluated the wearable with biceps-curl trials, refining sensor placement, calibration thresholds, and rep-detection logic.

Light-Responsive Alarm System (Project)

March 2026 - May 2026

University of Cincinnati, Cincinnati, OH

- Designed an ESP32-based embedded system with sensors, LCD display, and buzzer for responsive monitoring.
- Processed analog signals with a 12-bit ADC and applied averaging to reduce noise and stabilize sensor measurements.
- Implemented Wi-Fi time synchronization and designed a custom CAD enclosure in Fusion 360 for the prototype system.

Student Ambassador

August 2025 - May 2026

University of Cincinnati Blue Ash, Blue Ash, OH

- Represented UCBA as a campus leader by serving as the first point of contact for over 4,800 students and their families.
- Conducted campus tours and orientations for approximately 1,800+ prospective students and guests.
- Served as a Student Life liaison, coordinating events and digital outreach with staff to enhance student engagement.

Python & SQL Trainee (Project-Based)

May 2025 - August 2025

The Global Career Accelerator, Remote

- Analyzed industry datasets for Intel and The Grammys to evaluate hardware sustainability and platform user engagement.
- Applied Python (Pandas, NumPy) and SQL to clean, transform, and validate structured datasets in Jupyter Notebook.

Autonomous Parking Robot (Project)

January 2025 - April 2025

University of Cincinnati, Cincinnati, OH

- Designed and built an autonomous LEGO EV3 vehicle using iterative engineering design and testing.
- Programmed multi-sensor control (color, gyro, ultrasonic, touch) in LabVIEW for autonomous parking.

LEADERSHIP, ACTIVITIES, & SERVICE

STEMulates Robotics Volunteer

December 2025 - January 2026

The Gaskins Foundation, Cincinnati, OH

- Guided K-12 students through small hands-on robotics projects to introduce engineering problem-solving.

Vice President & Event Coordinator

October 2024 - May 2025

Latinos en Acción (LEA) Affinity Club, University of Cincinnati Blue Ash

- Led planning and execution of campus events, coordinating logistics, resources, and outreach to strengthen engagement.